

Guilherme Ribeiro Figueira

C++/C# Game Programmer (Junior)

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🌐 My Portfolio | ✉ gr.figueira@proton.me

Experience

Unity XR Developer - Curricular Internship

03/2023 - 08/2023

Glartek

Leiria, Portugal (Hybrid)

- Implemented an AR 3D model manipulation feature for **Microsoft Hololens 2** using Unity, C# and MRTK while collaborating with the web development and design teams, adding to the project's feature suit.
- Used **linear algebra**, **memoization techniques**, and **search algorithms** to optimize performance and ensure reliable processing of imported 3D model for the supported formats.
- Created **unit and integration tests** for scripts that used the **Zenject dependency injection** framework, improving test coverage and code reliability.
- Collaborated using an **AGILE** methodology with **GitLab** for version control.

Education

Nova School Of Science and Technology

Lisbon, Portugal

B.Sc. + M.Sc. in Computer Science and Engineering

2020 - 2025

- **Master Thesis:** Researched and developed a Unity-based framework that enables the abstract design of Substitutional Reality experiences, allowing them to be dynamically adapted to arbitrary physical spaces.
- Co-authored an additional paper [P1](#), on the topic.
- My implementation of **highly performant MR screen-space portals**, used in my research, is being ported into a **real-world museum application** at the *Portuguese National Tile Museum* in Lisbon, Portugal.

Skills

Programming Languages:

C++, C#, Java, Python, OCaml, HLSL

Engines & Frameworks:

Unreal Engine 5, Unity, FMOD, ImGui, SDL2

Graphics & Tools:

OpenGL, WebGL, Git, Shell, VIM, Docker

Creative Software:

Blender (Basics), Photoshop, DaVinci Resolve

Languages:

Portuguese (native), English (proficient)

Projects

The Shape of Silence - Unreal Engine 5 C++ Game

01/2024 - Present

Personal Project

- A narrative first person puzzle I started developing to sharpen my Unreal Engine C++ skills. I am working on this project solo, doing all the game design, writing, sound design, programming and more.
- So far I have developed an **extendable grid-based puzzle mechanic**, an **inventory subsystem** with support for 3D item inspections, and a non-linear **dialogue system** using my own **EasyDialogueEditor** tool to write the dialogue.

OpenGL 3D renderer

07/2025 - Present

Personal Project

- Used **C++** and **OpenGL** to implement a decoupled renderer, separating the rendering, application, shaders and settings.
- Finished features include: **3D fly-style camera**, **textured 3D objects**, a **fog setting**, and a **settings panel** added with **ImGui** to control camera, window and fog settings.
- On-going project, with the goal of eventually growing into my own small 3D game engine

Easy Dialogue Editor - C++ Desktop Application

03/2025 (3 weeks)

Personal Project

- A lightweight and platform agnostic **desktop program for editing game dialogue trees**. Created to solve a problem in a personal UE5 project.
- Written in **C++** with **ImGui**. It compiles to a **single small and portable executable** (~2.7 Mb) with no external dependencies, everything is statically linked.

The Natural's Descent - Unity Game

04/2024 - 06/2024

University Games and Simulations Course

- Action-adventure puzzle game developed in Unity as part of a **2-person team**.
- Designed and implemented a **procedurally generated cavern** using the **Wave Function Collapse** algorithm adapted to 3D, ensuring natural looking layouts with procedurally spawned enemies.
- Programmed multiple core gameplay mechanics such as: an **extendable sigil puzzle** component, a **ridable vehicle** and **FPS combat with AI animated humanoid enemies**.

Publications

P1 Towards Augmented XR Movement and Interaction Confidence across Substitutional Reality

Guilherme R. Figueira, Rui Nóbrega

International Conference on Graphics and Interaction (ICGI), 2025

Involvement

EXPO FCT 2025 Volunteer

30/04/2025

- Organized a **Mixed Reality** activity, building a short Unity MR game for the Meta Quest 3; demoed to high-school visitors and answered questions regarding the CS program.

DevGamm Lisbon 2024 Volunteer

15/11/2024

Replacement / Everything Volunteer

- Prepared rooms, guided attendees, and filled in across roles to keep sessions on schedule.